

Diverse Neural Photo Album Summarization

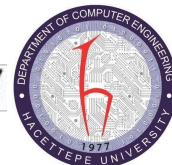
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9. International Conference on Image Processing Theory, Tools and Applications



HACETTEPE UNIVERSITY
COMPUTER VISION LAB



Motivation



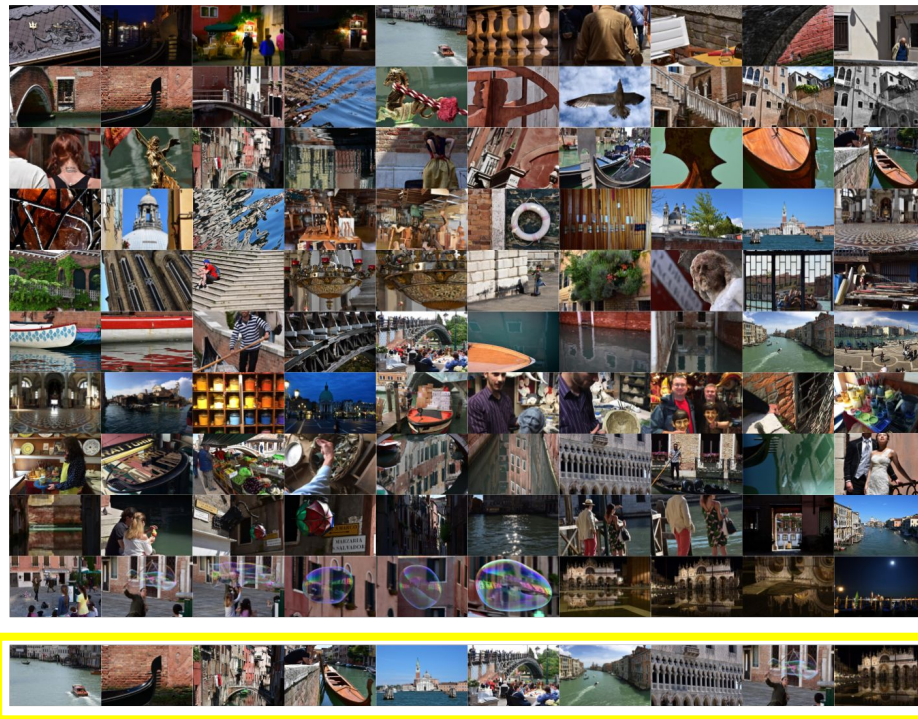
Problem Definition

The main objective is finding key images to represent a photo collection in the best way.



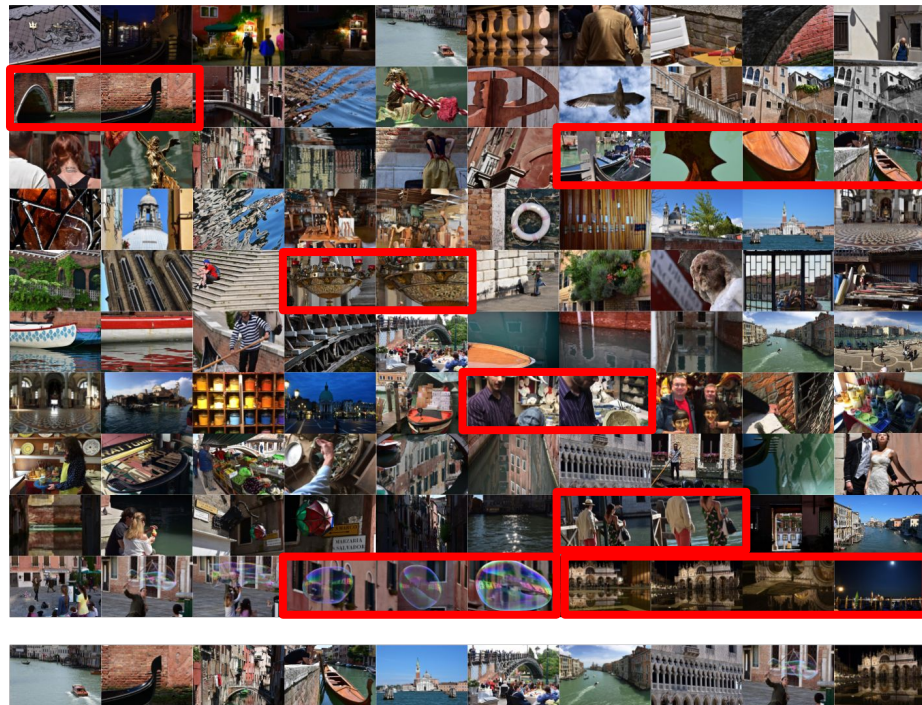
Problem Definition

The main objective is finding key images to represent a photo collection in the best way.



No Nearly similar images or same event

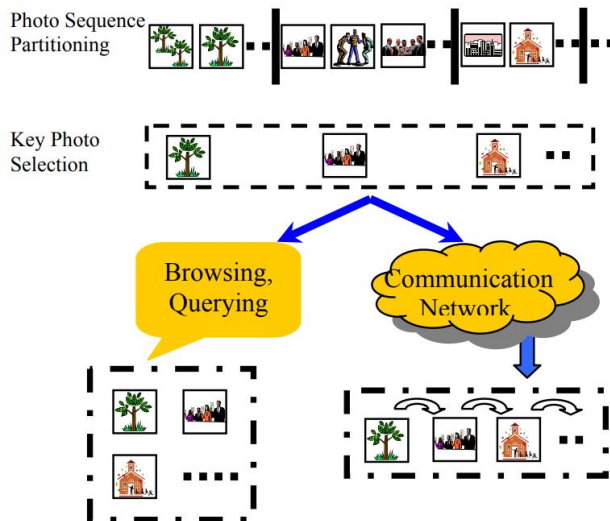
A representative summary should not contain similar images or images from same event.



How to summarize a photo album in a
smarter and representative way in
automatic manner?

Related Work

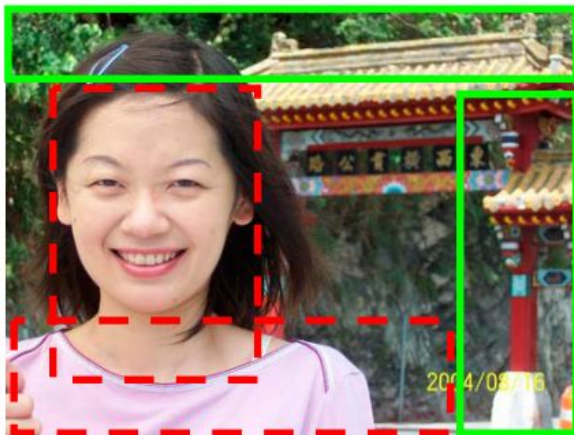
- Using color histograms and detected faces *
 - First, the photos are first partitioned according to time and then by image content
 - Second, key photos are selected from the partitions based on image content



* Jun Li, Joo Hwee Lim, and Qi Tian. Automatic summarization for personal digital photos. In Proc. of ICICS-PCM, volume 3. Citeseer, 2003

Related Work

- Person-centric method *
 - For each photograph, human faces, together with corresponding clothes and nearby regions are located
 - Extracting color histograms of these regions as the image content feature
 - Perform hierarchical clustering



(a)



(b)

* Cheng-Hung Li, Chih-Yi Chiu, Chun-Rong Huang, Chu-Song Chen, and Lee-Feng Chien. Image content clustering and summarization for photo collections. In 2006 IEEE International Conference on Multimedia and Expo, pages 1033–1036. IEEE, 2006

Related Work

- Combining hand-crafted features for encoding & random forest classifier *
 - Gathering a large dataset comprised of photo series distilled from personal photo albums
 - Color, lighting, deep features
 - Random forest classifier



* Huiwen Chang, Fisher Yu, Jue Wang, Douglas Ashley, and Adam Finkelstein. Automatic Triage for a Photo Series, ACM Transactions on Graphics (Proc. SIGGRAPH), July 2016

Related Work

- S-RNN: A recurrent neural network which learns a transition by skipping a lot of images in the sequences *



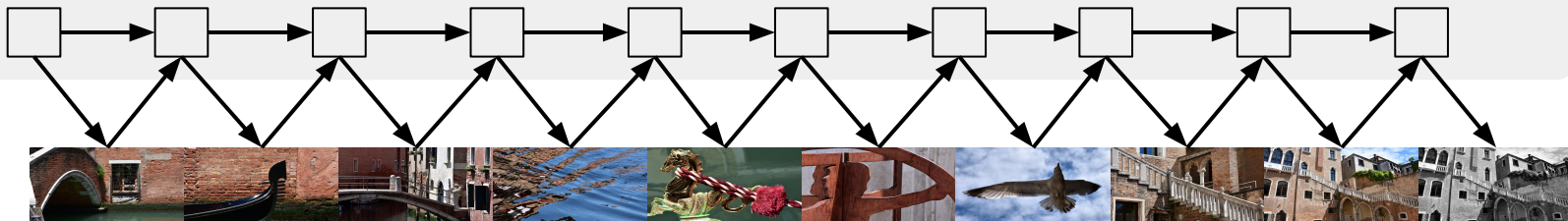
* Gunnar A Sigurdsson, Xinlei Chen, and Abhinav Gupta. Learning visual storylines with skipping recurrent neural networks. In European Conference on Computer Vision, pages 71–88. Springer, 2016

Our Approach

- Background & Baseline
 - Recurrent Neural Networks (RNN)
 - Skipping Recurrent Neural Networks (S-RNN)
- Proposed Method: Combined Discriminative Model (CDM)

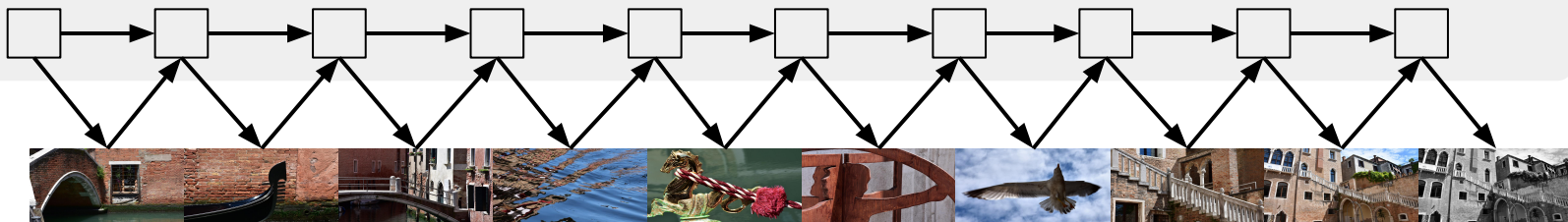
Recurrent Neural Networks (RNN)

RNN



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RNN

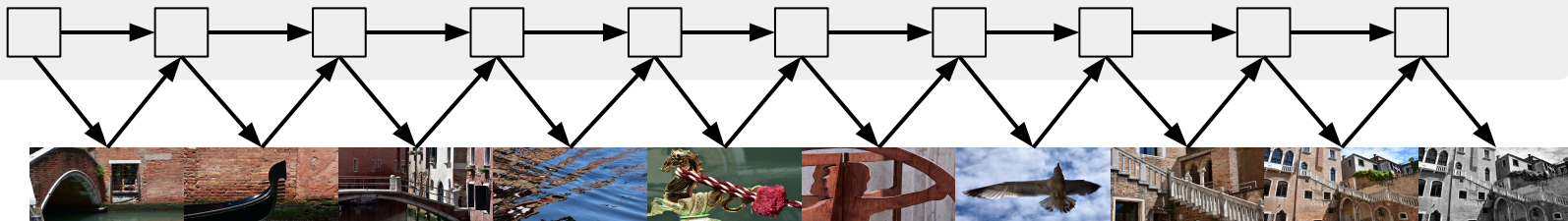


$$h^t = W_h h^{t-1} + W_x x^t$$

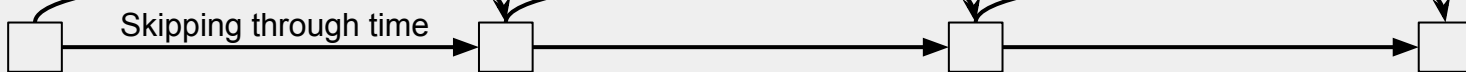
$$y^t = W_y h^t$$

Skipping Recurrent Neural Network (S-RNN)

RNN



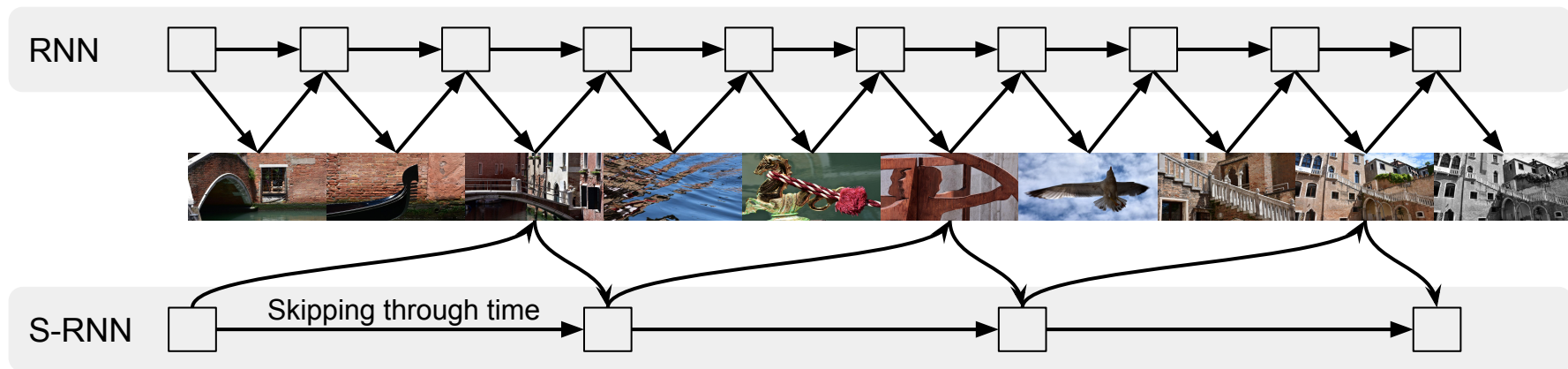
S-RNN



Summary



Skipping Recurrent Neural Network (S-RNN)



$$\mathcal{M}^* = \operatorname{argmax}_{\mathcal{M}} \log \sum_{\mathbf{z}_{1:N}} \left(\prod_n P(\mathbf{x}_{z_{n+1}} | \mathbf{x}_{\mathbf{z}_{1:n}}; \mathcal{M}) \right) P(\mathbf{z}_{1:N})$$

Generated summaries may contain similar images

Snowboarding

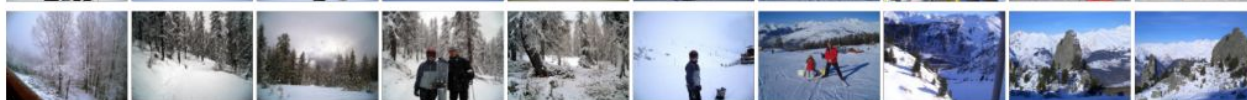
Person 1



Person 2



S-RNN



London

Person 1



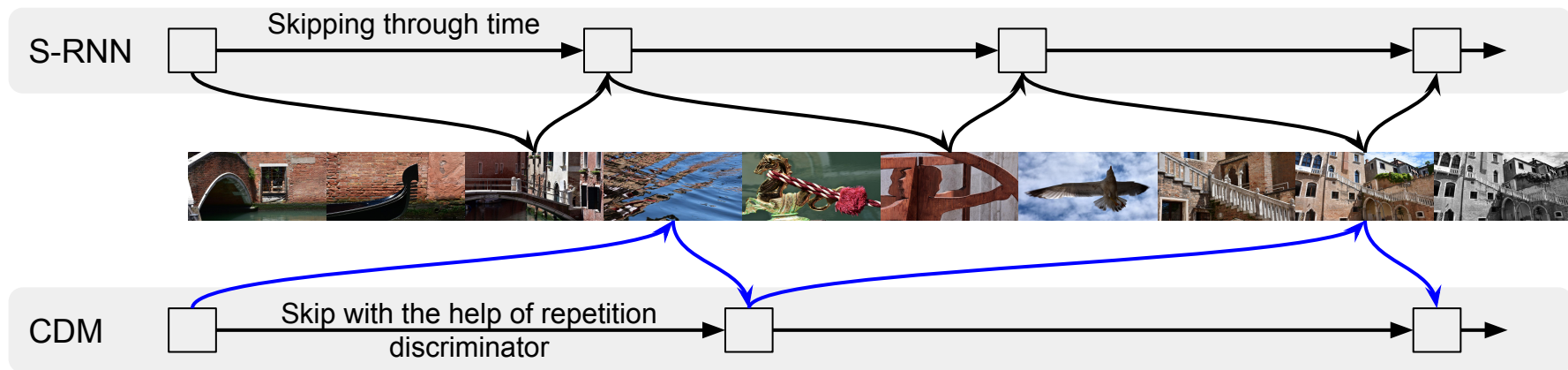
Person 2



S-RNN



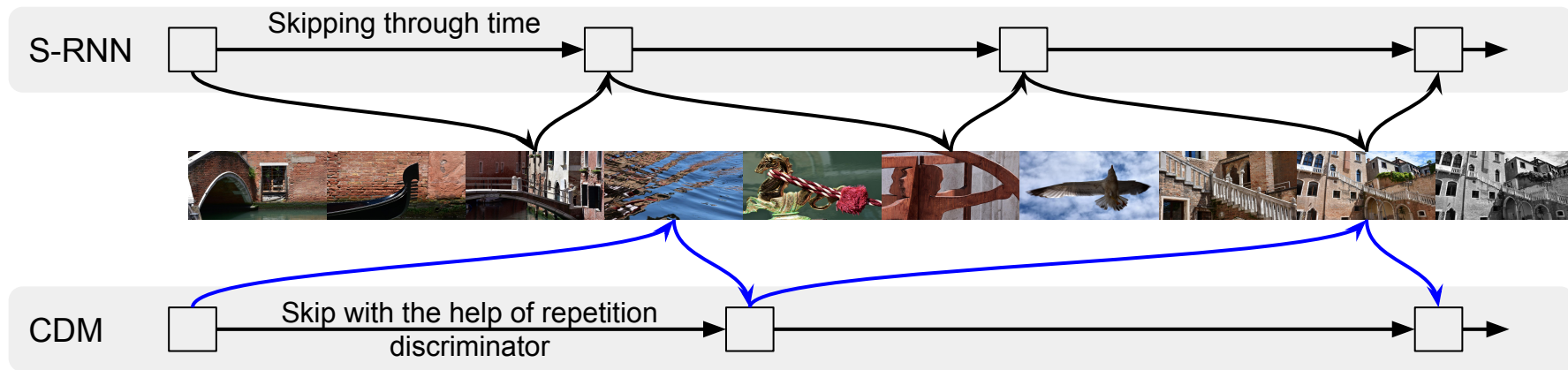
Combined Discriminative Model (CDM)



Summary



Combined Discriminative Model (CDM)



$$\mathcal{M}^* = \operatorname{argmax}_{\mathcal{M}} \left(\log \sum_{\mathbf{z}_{1:N}} \left(\prod_n P(\mathbf{x}_{z_{n+1}} | \mathbf{x}_{\mathbf{z}_{1:n}}; \mathcal{M}) \right) + \lambda s_{rep}(\mathbf{x}_{\mathbf{z}_{1:N}}) \right) P(\mathbf{z}_{1:N})$$

Repetition Score Function

1. Compute pairwise similarities in a fixed window of k photo.

$$d_i = \max_{j=i-k \dots i-1} (CosSim(y_j, y_i))$$

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3. Maximize ranking loss

$$L_{rep} = \sum_{\mathbf{y}_s \sim S-RNN(\mathbf{x})} \log \sigma(\beta - s_{rep}(\mathbf{y}_s))$$

Experimental Results

Experiments

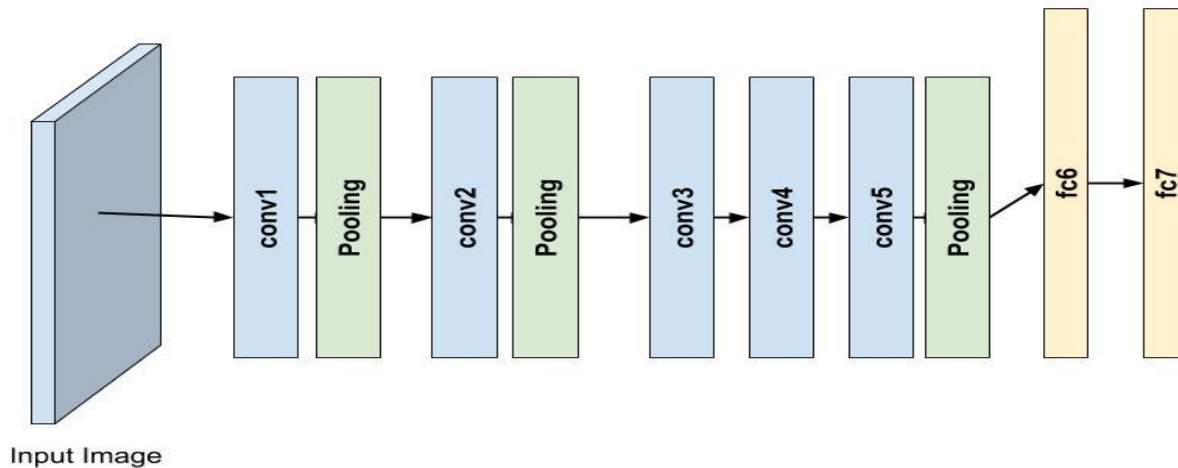
- Implementation and Training Details
 - Dataset*

Dataset	# of albums	Avg. # of images in the albums
Amsterdam	100	99
New York	1154	75
Paris	728	90
Tokyo	932	83
Venice	265	120

Experiments

- Implementation and Training Details
 - Feature Extraction

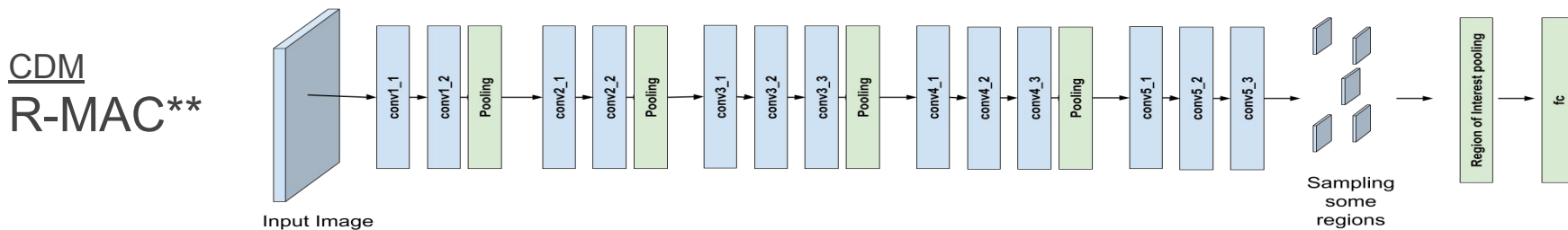
SRNN
Alexnet*



*Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In Advances in neural information processing systems, pages 1097–1105, 2012.

Experiments

- Implementation and Training Details
 - Feature Extraction



**Giorgos Tolias, Ronan Sifre, and Herve Jegou. Particular object retrieval with integral max-pooling of cnn activations. Proceedings of the International Conference on Learning Representations. 2016

Experiments

- Implementation and Training Details
 - Hyper parameters

Lambda : 1

Beta : 0.4

Window size k: 2

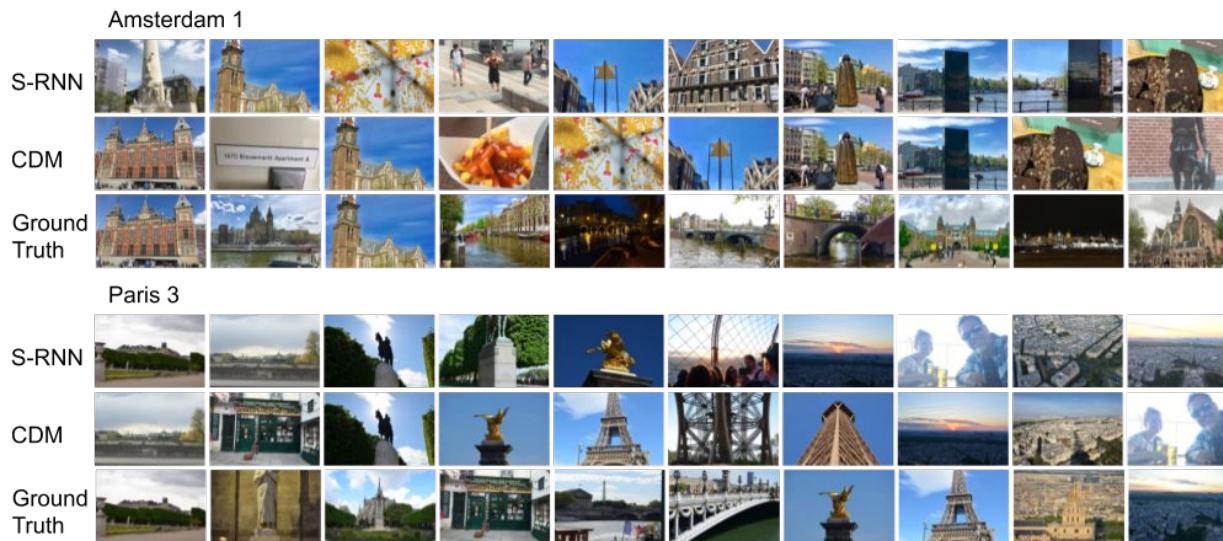
Learning rate of repetition function: 0.0001

Optimizer: SGD

Album length: 10

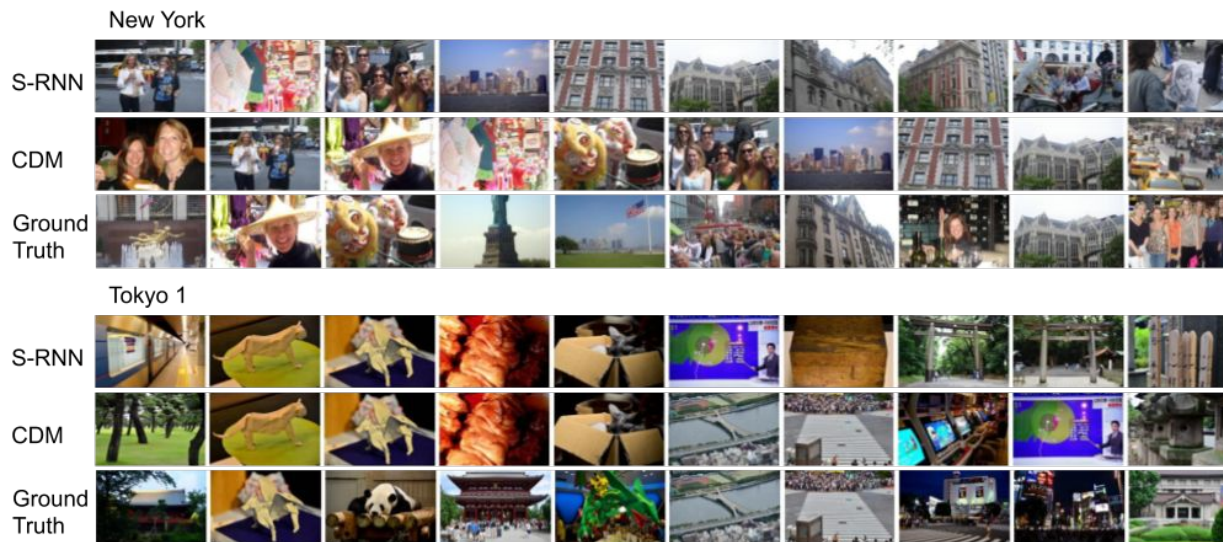
Experiments

- Analysis
 - Qualitative Analysis
 - Quantitative Results



Experiments

- Analysis
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Experiments

- Analysis
 - Qualitative Analysis
 - Quantitative Results
 - Diversity
 - Summarization

Photo Streams	Cosine Similarity		LPIPS [*]	
	S-RNN	CDM	S-RNN	CDM
Amsterdam 1	0.9936	0.9943	0.6579	0.6579
Amsterdam 2	0.9941	0.9947	0.6096	0.5779
Newyork	0.9937	0.9935	0.6780	0.6984
Paris 1	0.9923	0.9923	0.6344	0.6447
Paris 2	0.9929	0.9937	0.6161	0.6219
Paris 3	0.9920	0.9903	0.6590	0.6502
Paris 4	0.9947	0.9950	0.6095	0.5827
Paris 5	0.9947	0.9944	0.6105	0.6267
Tokyo 1	0.9910	0.9913	0.7307	0.7344
Tokyo 2	0.9934	0.9937	0.6873	0.6823
Venice 1	0.9900	0.9926	0.6941	0.7096
Venice 2	0.9909	0.9913	0.6947	0.6794

* Zhang, Richard, et al. "The unreasonable effectiveness of deep features as a perceptual metric." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2018.



Experiments

- Analysis
 - Qualitative Analysis
 - Quantitative Results
 - Diversity
 - Summarization

Photo Streams	F-score		V-ROUGE*	
	S-RNN	CDM	S-RNN	CDM
Amsterdam 1	0.0516	0.0525	0.3019	0.3030
Amsterdam 2	0.0854	0.0849	0.4350	0.4385
New York	0.1118	0.1116	0.4083	0.4053
Paris 1	0.0709	0.0709	0.3403	0.3548
Paris 2	0.1519	0.1480	0.5010	0.4976
Paris 3	0.0671	0.0695	0.3240	0.3279
Paris 4	0.1635	0.1479	0.3825	0.3695
Paris 5	0.0758	0.0783	0.4490	0.4543
Tokyo 1	0.0747	0.0789	0.4117	0.4213
Tokyo 2	0.0875	0.0823	0.4389	0.4302
Venice 1	0.0679	0.0782	0.3234	0.3338
Venice 2	0.0955	0.0956	0.3574	0.3571

Conclusion

A combined discriminative model to avoid repetition in summaries of photograph albums with a different image representation.

LPIPS

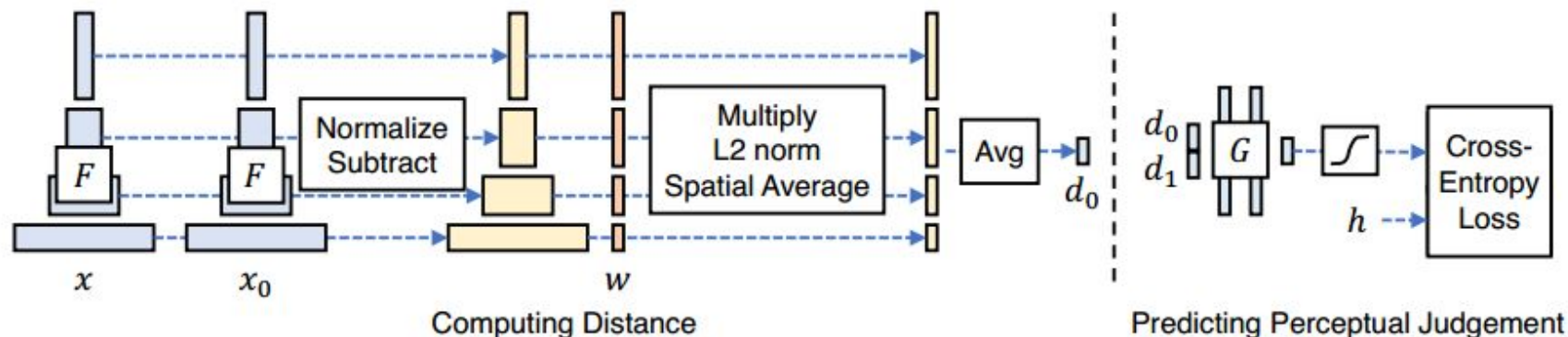
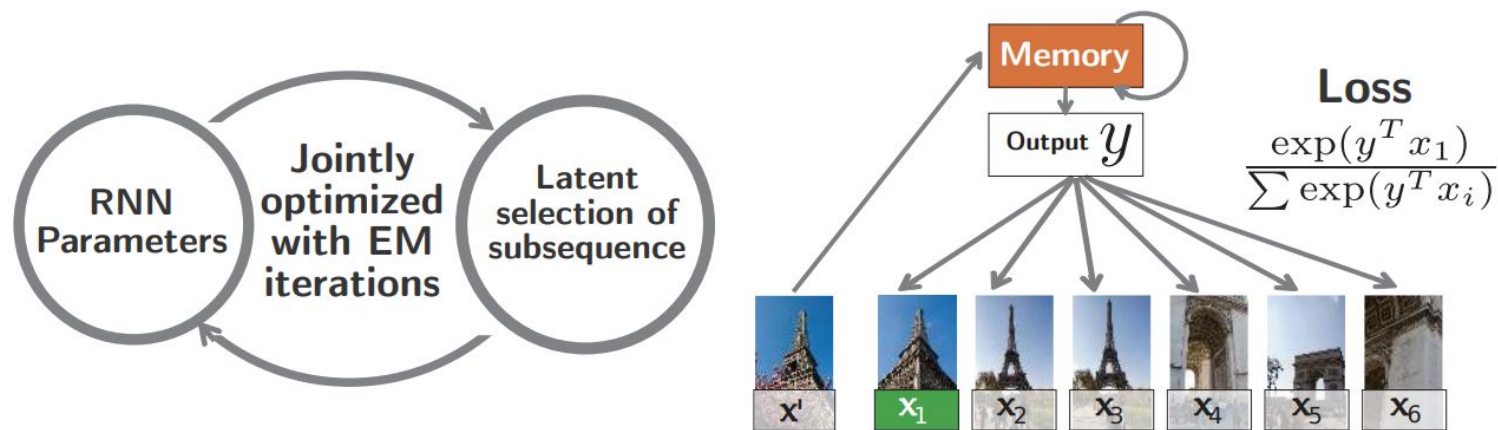


Figure 3: **Computing distance from a network** (Left) To compute a distance d_0 between two patches, x, x_0 , given a network $\mathcal{F}$, we first compute deep embeddings, normalize the activations in the channel dimension, scale each channel by vector w , and take the ℓ_2 distance. We then average across spatial dimension and across all layers. (Right) A small network $\mathcal{G}$ is trained to predict perceptual judgment h from distance pair (d_0, d_1) .

S-RNN Training



- *Train RNN over all possible subsequences*
- Rewritten as sequential update equations for E and M steps
- Fits in the RNN pipeline: *Infer next variable, then backprop*
- Prediction is *softmax over future images*
 - Images are represented as their fc7 features

V-Rouge